

System Modeling

System modeling is the process of developing abstract models of a system.
Each model presents a different view or perspective of the system.
Models are usually drawn with the Unified Modeling Language (UML).

Four Perspectives

External perspective: the context or environment of the system.

Interaction perspective: interactions between the system and its environment.

Structural perspective: the organisation of the system or its data.

Behavioral perspective: the dynamic behavior of the system and how it responds to events.

Use Case Diagrams

A use case represents a discrete task that involves external interaction with a system.

Actors can be people or other systems.

The include relationship always happens; the extend relationship happens only under a condition.

Sequence Diagrams

Sequence diagrams show the interactions between actors and objects during one use case. Objects and actors are listed along the top, each with a dotted lifeline. Interactions are drawn as annotated arrows.

State Diagrams

State diagrams model how a system reacts to internal and external events.

Event-driven modeling assumes a finite number of states.

A superstate groups several states and looks like a single state in a high-level model.